

When “\$1” Isn’t One: Stablecoin Discounts and Market Fragmentation

Abstract— Stablecoins have become central settlement instruments in crypto markets. However, their reliance on credible redemption and reserve transparency leaves spot pricing vulnerable when pegs come under stress. The March 2023 USDC de-peg offers a natural experiment to test whether cross-currency markets continue to satisfy the law of one price when the unit of account becomes uncertain. We study cross-quote dislocations and liquidity fragmentation using high-frequency BTC/USD and BTC/USDT data, with BTC/USDC where available. We quantify spot-spot deviations net of realistic transaction costs and find economically meaningful dislocations concentrated in the crisis regime. We then assess how stablecoin discounts transmit into BTC pricing and how liquidity fragments across quote currencies using tick-level price impact and illiquidity measures. Lastly, we model dislocation persistence with a regime-based mean-reversion framework and discuss how stronger reserve disclosure and redemption standards may reduce the severity and duration of stablecoin-driven inefficiencies.

I. INTRODUCTION

In this paper, we study stablecoin-driven fragmentation in cryptocurrency spot markets and its implications for pricing efficiency, liquidity, and execution risk under stress, using the IAQF March 1–21, 2023 UTC window that contains the USDC de-peg episode. The financial motivation is straightforward. Crypto spot markets often quote and clear trades in stablecoins that are intended to represent a dollar. When that representation is credible, stablecoin-quoted and USD-quoted markets should be tightly linked by arbitrage [5]. When credibility is questioned, the unit of account itself becomes risky, and cross-quote parity can break down even if nothing fundamental has changed about the underlying asset.

The March 2023 episode provides a clean stress test of this unit-of-account channel. USDC traded materially below one dollar during the Silicon Valley Bank shock, reaching roughly \$0.88 at its trough, before recovering as confidence in reserves and redemption was restored. This type of event is not only a market microstructure curiosity. It directly affects execution quality for institutional traders who rely on stablecoins for collateral, funding, and settlement, and it can create basis risk for hedges that assume stablecoin quotes are interchangeable with dollars. It also matters for market stability, since dislocations can coincide with liquidity withdrawal and slower normalization, precisely when market participants most need reliable pricing.

The regulatory motivation is that stablecoin policy is explicitly aimed at reducing these credibility shocks. The House Financial Services Committee’s section-by-section summary of the GENIUS Act describes a federal framework for payment stablecoins, including one-to-one reserve backing in specified

liquid assets and ongoing public reserve reporting subject to independent examination [13]. At the same time, stablecoin rails are moving closer to traditional settlement infrastructure, illustrated by Visa’s U.S. launch of USDC settlement for eligible issuer and acquirer partners [14]. Together, these developments raise a practical question for markets and policymakers, namely, if stablecoins are becoming settlement infrastructure, how *resilient* is cross-quote price formation and liquidity when stablecoin credibility is stressed?

Our empirical approach follows the competition questions and addresses them in a holistic manner. First, we measure one-minute law-of-one-price deviations for BTC across USD and stablecoin quotes and assess economic significance net of realistic transaction costs. Second, we examine how stablecoin premia and discounts transmit into BTC pricing and liquidity using standard microstructure measures, including price impact and illiquidity. Third, we quantify normalization dynamics with a regime-based mean-reversion framework, showing materially slower adjustment during stress, consistent with constrained arbitrage when the unit of account is questioned. We use these results to draw mechanism-based implications for execution and trading, for basis and liquidity risk management, and for policy design aimed at reducing stablecoin-driven dislocations and their persistence.

II. DATA AND METHODOLOGY

A. Data

We work with the IAQF sample window from March 1 to March 21, 2023 (UTC) at one-minute frequency, which gives 30,240 observations and covers the USDC de-peg episode around March 10 to 12. Table I summarizes the raw inputs. Our core dataset is a harmonized one-minute OHLCV panel from Binance.US and Coinbase for BTC quoted in USD and stablecoins, together with Binance.US stablecoin parity series such as USDC/USD and USDT/USD [8], [10].

To study liquidity and fragmentation, we use Binance tick trades from the public aggTrades archive and one-second klines, then aggregate these into one-minute microstructure measures so they line up with the pricing panel [9]. We also use Kraken and Bybit tick data aggregated to one minute as robustness checks, mainly to confirm that the patterns we document are not limited to a single venue. Two constraints are important for interpretation and are stated in Table I, wherein BTC/USDC starts trading on Binance on March 12, and Coinbase’s public candle endpoint returns identical BTC-USD and BTC-USDC candles, so BTC/USDC price formation results rely primarily on

Binance.US.

TABLE I
Data Sources (Mar 1–21, 2023)

#	Venue	Instrument(s)	Type	Res.	Size
1	Binance.US	BTC/USDT, USDC, USD, BUSD; USDT/USD, USDC/USD, USDC/USDT, BUSD/USDT	OHLCV	1-min	30,240 rows × 8 series
2	Coinbase	BTC-USD, BTC-USDT	OHLCV	1-min	~30,000 rows × 2 series
3	Binance (archive)	BTC/USDT (21d); BTC/USDC (10d, listed Mar 12)	Tick (aggTrades)	~ms	~150.6M (USDT); ~557K (USDC)
4	Binance (archive)	BTC/USDT, BTC/USDC	OHLCV	1-sec	~1.8M rows (USDT); ~864K (USDC)
5	Kraken	BTC/USD, USDC/USD, USDT/USD, USDC/USDT	Tick trades	~ms	~1M+ trades → 29,881 rows (1-min agg)
6	Bybit	BTC/USDT	Tick trades	~ms	~30M+ trades → 30,240 rows (1-min agg)

All data cover Mar 1–21, 2023 (UTC), centred on the USDC de-peg event (Mar 10–12). BTC/USDC listed on Binance Mar 12; Coinbase public API returns identical OHLCV for BTC-USD and BTC-USDC.

B. Methodology

Table II shows how these inputs map into the IAQF questions. For cross-currency basis, we construct log law-of-one-price wedges comparing BTC quoted in stablecoins to BTC quoted in USD after adjusting for the stablecoin’s USD value, then evaluate economic relevance using round-trip transaction-cost hurdles and hurdle exceedance rates. For stablecoin dynamics, we measure deviations from parity, split the sample into pre-crisis, crisis, recovery, and post regimes, and decompose basis variation into a stablecoin repricing component and a residual venue component. For liquidity and fragmentation, we compute tick-based market quality measures such as price impact, illiquidity, imbalance, and simple spread and depth proxies, and compare them across quote currencies and regimes. Finally, to quantify how quickly dislocations close, we estimate regime-dependent mean reversion of the residual wedge and report half-lives as a compact measure of arbitrage capacity under stress versus normal conditions. Further details on methodologies are detailed in later sections.

TABLE II
Data Application to IAQF Research Questions

#	Data Used	What We Computed	Output	Q
1	Binance.US + Coinbase 1-min OHLCV	Log LOP deviations; stablecoin FX dev. from par; realized vol.; volume share; OLS regressions on basis	30,240-row panel; Figs 1–5	Q1, Q2
2	Binance.US 1-min stablecoin FX pairs	USDT/USD, USDC/USD, USDC/USDT deviation series; regime classification	Regime labels applied to all analyses	Q2
3	Binance aggTrades (~150M rows)	Kyle’s Lambda; Amihud illiquidity; trade-based OBI; 1-min L2 panel	30,240-row L2 panel (USDT); 14,010-row (USDC); Figs 4–5	Q3
4	Binance 1-sec klines	Intra-minute realized variance; input to Kyle’s Lambda denominator	Merged into L2 panel	Q3
5	Kraken tick → 1-min	3-way BTC/USD venue wedges; wedge distribution by regime; cross-venue USDC/USD deviations	Figs K-1–K-3; Tables K-1, K-2	Q1, Q2
6	Bybit tick → 1-min (BTC/USDT)	Spread proxy, depth proxy, Kyle’s Lambda, OBI on offshore venue; stress vs. normal; lead-lag vs. USDC stress	Figs B-1–B-3; Table B-1	Q3

Q = IAQF research question addressed. L2 panel = combined liquidity/microstructure panel. Regime labels from Table II row 2 applied consistently across all downstream analyses.

C. Regime definition and assignment

We measure stablecoin premia and discounts as the log deviation of a stablecoin’s USD price from parity, expressed in bips as $10,000 \times \ln(S_t)$. Negative values indicate a discount to one dollar and positive values indicate a premium. We apply a single regime label throughout the paper so that stablecoin behavior, cross-quote wedges, liquidity measures, and persistence estimates are evaluated on the same event-time partition.

Our regime classification is event-driven and set ex ante. We use fixed calendar cutoffs that bracket the known chronology of the USDC de-peg, and that is consistent with standard event-study practice in which researchers define windows around externally identifiable news and compare behaviour before,

during, and after the event.

The relevant shock here is the SVB episode and Circle’s disclosure of reserve exposure. Concretely, we define pre-crisis as March 1 00:00 to March 9 23:59 UTC, crisis as March 10 00:00 to March 12 23:59, recovery as March 13 00:00 to March 15 23:59, and post as March 16 00:00 to March 21 23:59. Each one-minute timestamp is assigned by direct comparison to these fixed UTC boundaries, and the same boundaries are applied across all datasets to ensure consistency

III. LAW OF ONE PRICE AND MARKET FRAGMENTATION

Crypto spot markets increasingly use stablecoins as units of account and settlement rails, which improves transferability but concentrates pricing integrity on redemption credibility and reserve backing. The March 2023 USDC de-peg provides a stress episode (pre-dating U.S. federal stablecoin frameworks such as the GENIUS Act) to test whether BTC quoted in stablecoins continues to satisfy the law of one price against BTC/USD once the unit of account is questioned. We study BTC/USDT versus BTC/USD by (A) defining a log cross-quote basis and decomposing it into a venue-pricing leg and a stablecoin-FX leg, (B) establishing long-run parity via stationarity and cointegration of the constructed wedge, (C) quantifying regime-dependent persistence by mapping an AR(1) fit into OU mean-reversion speeds and half-lives, and (D) assessing economic tradability by imposing round-trip transaction-cost hurdles and evaluating hit rates of hurdle exceedance.

A. Decomposition of Cross-Currency Parity Deviations

A natural starting point is to define a parity deviation that compares BTC quoted in different units of account while adjusting for the stablecoin’s USD value.

Let $P_t^{BTC/USDT}$ and $P_t^{BTC/USD}$ denote BTC prices quoted in USDT and USD at time t , and let $S_t^{USDT/USD}$ denote the USDT/USD exchange rate. The residual parity deviation is defined as:

$$b_t = \ln(P_t^{BTC/USDT}) - \ln(P_t^{BTC/USD}) - \ln(S_t^{USDT/USD})$$

Under frictionless parity and a stable unit of account, $b_t = 0$. Deviations reflect a wedge that can arise from either fragmentation across order books and venues, or stablecoin discounting that distorts the quote currency.

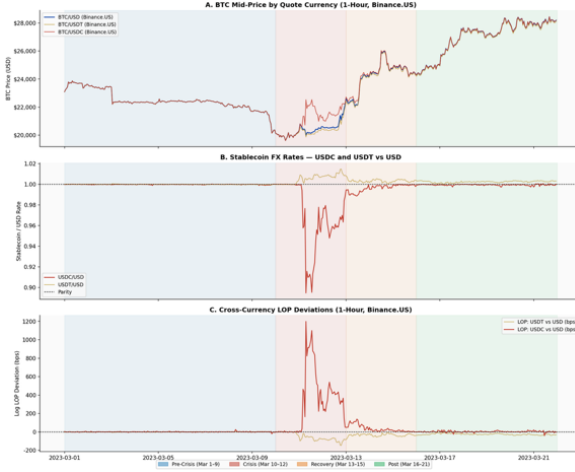


Fig. 1 plots BTC/USD, BTC/USDT, and the implied parity deviation over the full sample, highlighting the March 2023 USDC de-peg window. Deviations are tightly centered around zero in pre-crisis periods but widen sharply during the de-peg and remain elevated for extended intervals. At the peak of the episode, the BTC/USDC versus BTC/USD deviation reaches 1,324 basis points, roughly two orders of magnitude above typical pre-crisis levels. To attribute deviations, we use a two-leg decomposition:

$$x_t = \ln(P_t^{BTC/USDT}) - \ln(P_t^{BTC/USD}),$$

$$y_t = \ln(S_t^{USDT/USD}), b_t = x_t - y_t.$$

The contribution of each component is evaluated using the variance identity:

$$\text{Var}(b) = \text{Var}(x) + \text{Var}(y) - 2 \text{Cov}(x, y)$$

The covariance term captures interaction effects. If the two legs were weakly related, the interaction contribution would be small. Empirically, the interaction term is economically important and becomes most pronounced during the stress window.

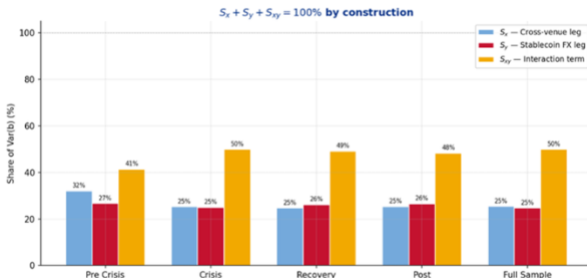


Fig. 2. Variance Attribution of Basis by Regime

Fig. 2 reports the variance attribution by regime. During the USDC de-peg episode, the correlation between x_t and y_t reaches -0.997 , and the interaction share accounts for approximately 49.9% of total basis variance. This indicates that stablecoin repricing and cross-venue quote dispersion moved in a tightly coupled manner during stress, amplifying the observed parity deviations.



Fig. 3. Basis Risk Decomposition — BTC/USDT vs BTC/USD (Binance.US, 1-min)

Fig. 3 plots the time series of x_t , y_t , and b_t , illustrating that large crisis-period deviations reflect the joint behavior of both legs rather than a single isolated channel.

The decomposition therefore reveals that crisis-period deviations are not driven by isolated market frictions. Instead, stablecoin discounting and cross-venue pricing respond jointly, producing amplified parity breakdowns

B. Long-Run Parity and Stationarity of the Basis Wedge

Before interpreting these deviations as structural inefficiencies, it is necessary to establish their time-series properties. Augmented Dickey - Fuller tests (Dickey & Fuller, 1979) indicate that BTC price levels exhibit a unit root, consistent with a non-stationary price process [3]. In contrast, both stablecoin FX deviations and the constructed LOP wedge b_t are stationary. Engle-Granger cointegration tests (Engle & Granger, 1987) further confirm the existence of a long-run equilibrium relationship between BTC/USD and BTC/USDT once the FX leg is included [2].

TABLE III
Augmented Dickey-Fuller Unit Root Tests

Series	ADF stat.	p-val.	Conclusion (5%)	Order
BTC/USD (Binance.US)	0.271	0.976	Fail to reject	I(1)
BTC/USDT (Binance.US)	0.297	0.977	Fail to reject	I(1)
BTC/USDC (Binance.US)	0.031	0.961	Fail to reject	I(1)
USDC/USD rate	-3.519	0.008	Reject (stationary)	I(0)
USDT/USD rate	-3.223	0.019	Reject (stationary)	I(0)
LOP wedge: USDT vs. USD	-3.179	0.021	Reject (stationary)	I(0)
LOP wedge: USDC vs. USD	-3.365	0.012	Reject (stationary)	I(0)

ADF test with constant; lag order by AIC. Critical value at 5%: -2.86 . I(1): unit root; I(0): stationary.

These results imply that cross-currency pricing satisfies parity in the long run. Deviations represent mean-reverting wedges rather than drifting spreads. The stationarity of b_t is essential: it confirms that the large crisis deviations observed in Figure 1 are temporary dislocations rather than structural divergence.

C. Breakdown of Mean Reversion in Crisis Regimes

Although parity holds in the long run, the magnitude and persistence of deviations vary sharply across regimes. In pre-crisis conditions, the average absolute LOP deviation remains approximately 1–2 basis points. During the crisis window, deviations increase to triple-digit basis points and remain elevated for extended intervals. We estimate a first-order autoregressive process:

$$b_{t+1} = \alpha + \phi b_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \text{i.i.d. } (0, \sigma^2)$$

To interpret persistence in economically meaningful time units, we map the discrete-time coefficient ϕ to a continuous-time Ornstein-Uhlenbeck (OU) process:

$$db_t = \kappa(\mu - b_t) dt + \sigma dW_t,$$

where $\kappa > 0$ is the speed of mean reversion [1]. Under standard discretization with sampling interval Δt , the OU parameters and associated half life of a shock is:

$$\kappa = -\frac{\ln(\phi)}{\Delta t}, \quad \text{Half-life} = \frac{\ln(2)}{\kappa}.$$

In normal regimes, deviations revert rapidly toward zero. During the crisis window, the implied half-life extends to approximately 10 hours, indicating substantially slower parity enforcement. The increase in persistence coincides with the increase in magnitude, suggesting that arbitrage capacity weakens precisely when deviations are largest.

TABLE IV
Regime AR(1)/OU Parameter Estimates

Regime	N	ϕ	κ	$t_{1/2}$	μ	σ	ADF
Pre (Mar 1–9)	12,960	0.807	0.214	3.2	−0.3	2.2	S
Crisis (Mar 10–12)	4,320	0.999	0.001	602.7	−140.3	3.8	NS
Recov. (Mar 13–15)	4,320	0.982	0.018	37.7	−68.6	5.4	S
Post (Mar 16–21)	8,640	0.964	0.037	18.7	−51.1	4.4	S

ϕ : AR(1) coef.; κ : OU speed (min^{-1}); $t_{1/2}$: half-life (min); μ , σ in bps; S/NS: stationary/non-stationary (ADF).

Table IV shows the numbers. Pre-crisis, $\phi = 0.8074$ implies a half-life of 3.2 minutes, and the unconditional mean remains near zero. Parity is enforced rapidly. During the crisis window, ϕ rises to 0.9989, implying a half-life of approximately 602.7 minutes (roughly 10 hours). The process mean shifts sharply negative (−140 bps), and the ADF test fails to reject a unit root within this subsample (Table III). Mean reversion effectively collapses.

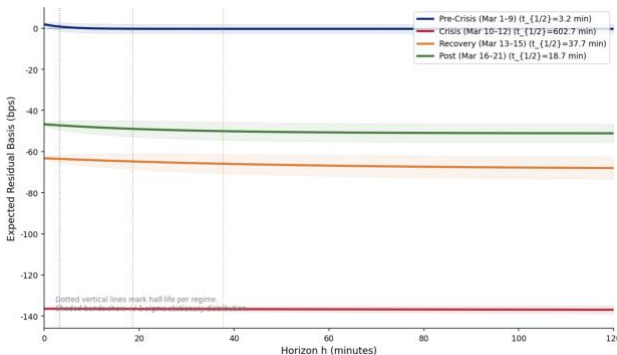


Figure 4: OU Impulse Response to a 1-Sigma Shock by Regime

Figure 4 plots the OU impulse response to a one-standard-deviation shock by regime. In pre-crisis conditions, shocks decay toward equilibrium within minutes, whereas in the crisis regime the impulse response remains close to flat over a two-hour horizon, consistent with the high persistence in Table III.

D. Implementability of Arbitrage under Transaction Costs

The preceding results establish that deviations are mean-reverting in normal conditions but become larger and substantially more persistent during the crisis regime. The remaining question for this component, is whether these dislocations were economically tradable once realistic frictions are imposed. The remaining question is whether these deviations were economically tradable once realistic frictions are imposed.

We therefore compare the observed basis to scenario-specific round-trip cost hurdles (Table V), which define a no-trade band. An arbitrage trade is considered implementable only when $|b_t|$ exceeds the relevant hurdle. We summarize implementability using the hit rate, defined as the fraction of one-minute intervals in which the basis clears the hurdle (Table IV).

Using this criterion, Table VI shows that in the pre-crisis regime the basis remains largely within the no-trade band across scenarios, consistent with low hit rates and limited

TABLE V
Round-Trip Arbitrage Cost Summary

Scenario	Fixed (bps)	Withdrawal (bps)	Total Hurdle
Binance.US Only (zero-fee promo)	10.00	0.00	10.00 bps
Cross-Exchange Retail (CB + BinUS)	70.00	1.25	71.25 bps
Cross-Exchange Inst. (>\$1M/mo.)	25.00	1.25	26.25 bps

Fixed cost includes maker/taker fees. Withdrawal costs are per-leg estimates. Total hurdle = minimum basis required for profitable round-trip.

implementable opportunity. During the crisis regime, the basis frequently breaches the cost hurdles, implying materially higher hit rates and indicating that the dislocations were often large enough to be actionable after costs. This timing supports a unit-of-account interpretation: when stablecoins trade away from parity, BTC prices quoted in stablecoins can embed premia or discounts relative to USD-quoted markets, and these premia/discounts interact with venue segmentation and liquidity conditions.

This motivates and serves as the foundation of the next part of our analysis (Q2–Q3), where we examine in extricate detail, the stablecoin premium/discount dynamics directly and relate them to cross-quote dislocations and liquidity fragmentation across quote currencies and venues.

TABLE VI
Arbitrage Hit Rate (%) by Scenario and Regime

Scenario	Pre	Crisis	Recov.	Post
Binance.US Only	0.4	70.3	100.0	98.9
Cross-Exch. Retail	0.0	62.8	40.9	6.3
Cross-Exch. Inst.	0.0	68.8	96.5	90.8

Hit rate = fraction of 1-min intervals where basis exceeds round-trip cost hurdle.

IV. STABLECOIN DYNAMICS

Stablecoins function as the effective unit of account for much of crypto spot trading, so their premium/discount behavior is not ancillary, it is the effective mechanism that turns a cross-quote comparison into a meaningful “law of one price” test. From previously establishing that cross-quote dislocations widened sharply during the USDC stress-event, it became meaningfully economical after transaction costs. As a result, it is now worth considering how stablecoin premia/discounts vary across exchanges and regimes, and what this implies for confidence under a forthcoming U.S. framework.

A. Regime Patterns in Stablecoin Deviations

To address how stablecoin premia and discounts vary across regimes, we start with Binance.US because it provides full one-minute coverage and the stablecoin parity series is used throughout our cross-quote wedge construction. The regime shift is concentrated in USDC.

In Table VII, Binance.US USDC/USD moves from a pre-crisis median of 0.0 bps to a crisis median of -296.3 bps, with a pronounced deterioration in the left tail ($P5 = -958.5$ bps; minimum = -1469.9 bps). These crisis outcomes are economically large and represent a direct unit-of-account distortion and represents market inefficiencies. USDT behaves differently over the same window. Binance.US USDT/USD remains comparatively stable and shifts mildly positive in the crisis regime (median approx. 60.8 bps; P95 approx. 113.4 bps). This contrast establishes the key regime and reinforces the conclusion from the previous section, that is, stablecoin discounts are state-dependent, and the stress episode is primarily a USDC-specific repricing rather than a uniform “stablecoin dollar” repricing. Table IX reports the USDC-USDT differential which has a crisis median of -377.7 bps.

TABLE VII
Stablecoin Premia/Discounts by Regime and Venue (bps)

Venue / Pair	Regime	N	P5	Median	P95	Min
Binance.US USDC/USD						
	Pre-Crisis	12,960	-5.0	0.0	1.0	-50.1
	Crisis	4,320	-958.5	-296.3	1.0	-1469.9
	Recovery	4,320	-104.5	-14.0	-1.0	-510.8
	Post	8,640	-14.0	-4.0	5.0	-35.1
Binance.US USDT/USD						
	Pre-Crisis	12,960	-2.0	0.0	3.0	-6.0
	Crisis	4,320	-1.0	60.8	113.4	-5.0
	Recovery	4,320	13.0	31.9	53.9	0.0
	Post	8,640	9.0	27.0	35.9	0.0
Kraken USDC/USD						
	Pre-Crisis	6,943	-1.0	0.0	0.0	-10.0
	Crisis	3,766	-947.5	-325.7	0.0	-1346.7
	Recovery	3,778	-94.4	-8.0	-4.0	-121.7
	Post	6,307	-21.7	-2.0	0.0	-32.1
Kraken USDT/USD						
	Pre-Crisis	12,422	-2.0	0.0	2.0	-5.0
	Crisis	4,319	-1.0	62.8	127.2	-2.0
	Recovery	4,318	20.0	38.9	62.8	11.0
	Post	8,605	11.2	25.0	33.3	2.0
Kraken USDC/USDT						
	Pre-Crisis	6,698	-2.0	0.0	2.0	-5.0
	Crisis	3,858	-998.2	-439.5	0.0	-1357.1
	Recovery	3,696	-134.1	-50.1	-28.0	-168.4
	Post	6,065	-38.1	-29.0	-19.0	-42.1

All values in bps. Log-deviation = $\ln(S_t) \times 10,000$. Negative = stablecoin at discount to USD par. USDC/USDT: negative = USDC at discount to USDT.

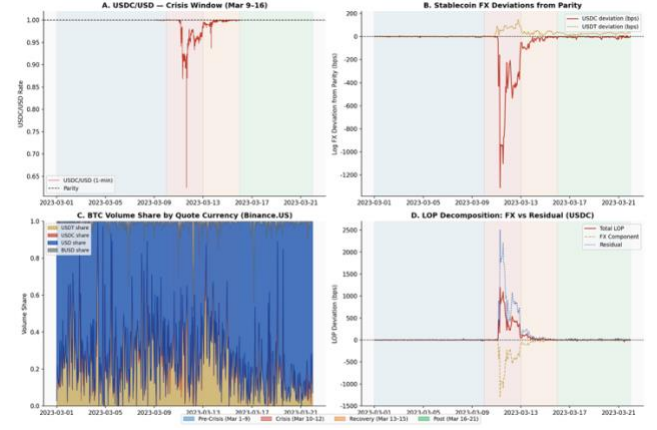


Figure 5: Stablecoin FX deviations, volume shares, and LOP decomposition (Binance.US; Mar 1–21, 2023). Panels: (A) USDC/USD in crisis window; (B) log FX deviations from parity for USDC and USDT; (C) BTC volume share by quote currency; (D) LOP decomposition for the USDC wedge into FX component vs residual.

Taken together, Figure 5 and Tables VII/IX show that premium/discount behaviour is strongly regime-dependent and that the stress episode is concentrated in USDC rather than a uniform repricing of all stablecoin units. The next question is

TABLE VIII
Cross-Venue USDC/USD Confidence Shock: Binance.US vs. Kraken

Regime	N	Corr(BinUS, Kraken)	BinUS Var (bps ²)	Kraken Var (bps ²)
Pre-Crisis	6,943	-0.030	22.4	0.3
Crisis	3,766	0.908	103,781.9	100,658.7
Recovery	3,778	0.763	1,514.7	906.2
Post	6,307	0.078	34.5	55.3

Crisis correlation of 0.908 and $\sim 5,000\times$ variance expansion confirm a market-wide systemic event, not venue-specific noise.

whether this behaviour is idiosyncratic to a single exchange or reflects a market-wide confidence shock.

B. Cross-Venue Validation of Stablecoin confidence

To address the cross-exchange question that still holds, we compare USDC/USD and USDT/USD on Binance.US and Kraken over the same regime partitions. Figure 6 shows that USDC/USD deviations co-move closely across the two venues during the crisis window, with a similar drawdown and recovery path.

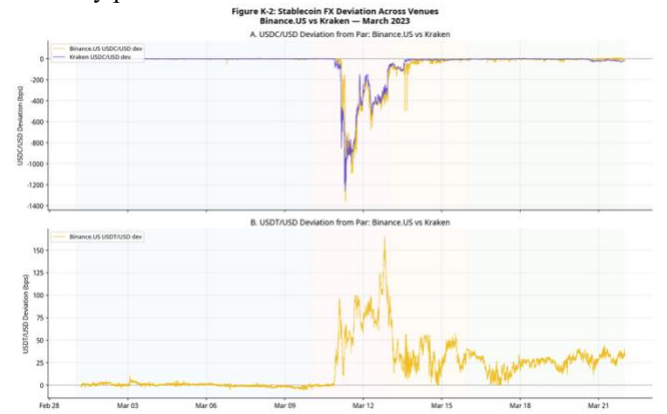


Figure 6: Stablecoin FX Deviations across Venues Binance.US vs Kraken during March 2023

The distributional magnitudes for USDC are similar across venues during the crisis regime. Table VII showcases the fact that Kraken USDC/USD has a crisis median of -325.7 bps,

with $P5 = -947.5$ bps and minimum = -1346.7 bps, and that is closely aligned with Binance.US. USDT remains relatively stable on Kraken as well (crisis median approx. 62.8 bps; P95 approx. 127.2 bps).

TABLE IX
USDC vs. USDT Relative Confidence (Binance.US, bps)

Regime	N	Median	P5	P95	Min
Pre-Crisis	12,960	-1.0	-5.0	2.0	-51.1
Crisis	4,320	-377.7	-1013.5	2.0	-1525.8
Recovery	4,320	-55.9	-138.5	-28.0	-537.8
Post	8,640	-30.0	-43.9	-15.0	-60.0

Series = $\log_{10}(\text{usdc_dev} / \text{usdt_dev})$. Negative = USDC at discount to USDT. Crisis median of -377.7 bps confirms market confidence loss specific to USDC; USDT acted as relative safe haven.

This cross-venue synchronization supports the interpretation that the de-peg reflected a market-wide confidence shock rather than an idiosyncratic single-order-book anomaly, which will be explored in greater detail in the next section. We next use these same objects to separate “market-wide confidence” from “relative confidence across stablecoins” more explicitly.

C. Transmission mechanism into cross-quote pricing

Building on the previous section’s evidence that cross-quote wedges became economically meaningful during the USDC stress window, the next step is to clarify the transmission channel. The key point is that a stablecoin discount changes the unit of account. When a “dollar” stablecoin trades below one dollar, assets quoted in that stablecoin can look mechanically expensive relative to their USD-quoted prices, even if underlying fundamentals are unchanged.

Figure 5 links directly to the dislocations documented earlier. Panel 5-B shows USDC’s discount opening sharply in the crisis regime, while Panel D decomposes the associated cross-quote deviation into an FX component and a residual component. The FX component is simply the stablecoin’s own deviation from parity, which mechanically shifts BTC/USDC relative to BTC/USD when USDC trades below one dollar. The residual component is what remains after stripping out this mechanical effect; formally, it is the wedge:

$$\ln(P^{BTC/USDC}) - \ln(P^{BTC/USD}) - \ln(S^{USDC/USD})$$

The residual, itself, captures quote-book dispersion and segmentation not explained by the stablecoin discount itself. During the stress window, the FX-like component turns strongly negative, consistent with a credibility shock to the unit of account, and the residual component spikes at the same time, indicating that BTC pricing across quote currencies also fragments when the stablecoin comes under pressure. In other words, the de-peg does not arrive as a clean, isolated FX event. It coincides with wider quote dispersion, so stablecoin repricing and market segmentation reinforce each other, exactly when execution conditions matter most.

D. Market-wide confidence shock versus relative confidence across stablecoins

We then examine whether the stablecoin discount was localized to one venue or reflected a broader confidence shock. Figure 6 makes the cross-venue picture clear in many senses, USDC/USD deviations on Binance.US and Kraken move

together through the stress window, with a similar drawdown and recovery path. The trough magnitudes are comparable, with USDC trading roughly 1,200 to 1,400 bps below parity on both venues, which is more consistent with a market-wide repricing than with a single order book behaving idiosyncratically.

This will be explored in greater detail in the next component. The de-pegging episode also reveals heterogeneity within stablecoins. On Kraken, USDC/USDT trades at a large crisis discount (median approx. -439.5 bps; P5 approx. -998.2 bps). On Binance.US, the USDC–USDT differential shows a crisis median of -377.7 bps, referring to Table IX. Taken together, the repricing is *concentrated* in USDC, with USDT priced at a relative premium during the same stress window, as noted earlier.

E. Confidence Implications

In market terms, “confidence” is not an abstract concept. It shows up in how deep the discount gets, how heavy the left tail becomes, whether the discount moves together across venues, and whether markets start ranking stablecoins against each other, as seen in Tables VII–IX. When confidence breaks, the stablecoin stops behaving like a clean numeraire, so a BTC price quoted in “stablecoin dollars” can mechanically diverge from BTC/USD even before we talk about trading frictions.

This is consistent with the stablecoin stress mechanism, articulated by the Federal Reserve, where secondary-market prices can deviate materially from par when redemption and reserve access become uncertain, and where primary and secondary channels can behave differently under stress [12]. Once the unit of account is under pressure, any existing segmentation and limits to arbitrage become more binding, which is consistent with the broader evidence in crypto markets that price gaps can persist when capital, transfers, and inter-venue arbitrage are constrained.

This is also why stablecoin regulation matters for the patterns we document. The House Financial Services Committee’s section-by-section summary of the GENIUS Act describes a framework aimed at strengthening redemption credibility and reserve assurance for payment stablecoins, including one-to-one reserve backing in specified liquid assets and recurring public reserve reporting subject to independent examination. If these mechanisms improve perceived redemption reliability in stress, the natural prediction is a smaller and shorter-lived left-tail discount distribution, and therefore a weaker unit-of-account distortion feeding into cross-quote dislocations.

Keeping this in mind, we move forward in a duo-fold manner. First, we check whether the USDC confidence shock shows up in market quality, by comparing spreads, depth proxies, volatility, and price impact for BTC quoted in USD versus stablecoins across venues and regimes. Second, we connect the mechanics back to policy by using our OU half-life estimates as a simple yardstick for how quickly pricing gaps close under stress. If regulation improves reserve transparency and redemption credibility, the natural expectation is smaller dislocations, faster normalization, and less liquidity fragmentation, which matters directly for execution and hedging.

V. LIQUIDITY AND FRAGMENTATION

The stablecoin results above show that the unit of account can become unstable and that the confidence shock propagates across venues. The natural follow-on question is whether market quality deteriorates at the same time, and whether that deterioration differs systematically across quote currencies. We answer this by comparing spreads, depth proxies, volatility, and two standard liquidity measures, price impact (Kyle’s lambda) and illiquidity (Amihud), for BTC quoted in USD versus stablecoins, using one-minute aggregates aligned to the same regime labels [6] [7].

A. Metrics

We work at a one-minute frequency so pricing wedges and market quality measures can be compared on the same clock. Our goal is to fundamentally quantify how costly it becomes to trade, and how much price moves per unit of trading activity, across quote currencies and regimes. We introduce these as follows.

Price impact (Kyle’s lambda). Kyle’s (1985) framework measures how much price changes per unit of signed order flow. For each minute t , let Δp_t be the log price change and q_t be signed volume, as shown:

$$\Delta p_t = \ln(P_t^{\text{close}}) - \ln(P_t^{\text{open}}), \quad q_t = \sum_{i \in t} s_i v_i,$$

Where, v_i is trade size and $s_i \in +1, -1$ indicating buyer-initiated versus seller-initiated trades. A simple per-minute impact proxy is shown as:

$$\lambda_t = \frac{|\Delta p_t|}{|q_t| + \epsilon},$$

With $\epsilon > 0$, as a small constant for numerical stability. A higher λ_t means that prices move more for a given amount of net trading pressure, which indicates thinner liquidity and poorer execution conditions.

The illiquidity metric of Amihud’s measure captures return per unit volume. At a one-minute frequencies:

$$ILLIQ_t = \frac{|r_t|}{Vol_t}$$

Where, we have r_t is the one-minute return and Vol_t is one-minute volume. Larger $ILLIQ_t$ implies larger price moves per unit volume. These were computed through proxies of:

$$SpreadProxy_t = \frac{High_t - Low_t}{Close_t} \times 10,000$$

$$DepthProxy_t = \frac{Volume_t}{SpreadProxy_t}$$

B. Quote-Currency Fragmentation: USD vs Stablecoins

With the aforementioned metrics in place, we first test whether liquidity differs systematically across quote currencies on the same venue, holding exchange-specific features fixed. The clearest comparison is between BTC/USDT and BTC/USDC on Binance, where both pairs are observed at one-minute frequency and the microstructure panel is computed from tick trades

aggregated into the same regime labels (Table II, Rows 3 - 4). A practical caveat matters for interpretation, that should be noted is that BTC/USDC is listed on March 12, 2023, which coincides with the peak of the USDC de-peg, noting Table II, on BTC/USDC). This means the USDC-quoted market for Binance is literally “born into crisis,” so we treat it as stress-conditional evidence on market quality rather than a pure comparative exercise.

Even with the listing caveat for BTC/USDC, the liquidity gap across quote currencies is hard to ignore. Around the de-peg episode, trading BTC in USDC looks meaningfully “thinner” than trading BTC in USDT. Quantitatively, Kyle’s lambda for BTC/USDC is about 64 times larger than for BTC/USDT, and Amihud illiquidity is about 4.4 times larger, as shown in Figure 7. In plain terms, that means two things, that is, a given amount of net buying or selling pressure moves the BTC/USDC price much more, and a given amount of traded volume is associated with larger price changes. Both point to worse execution conditions when the market is quoting in the stressed stablecoin.

BTC/USDT shows a fairly stable impact profile through the sample days, while BTC/USDC, from its March 12 listing onward, is much more erratic with pronounced spikes. The regime boxplots make the point cleaner, and more noticeable, with BTC/USDT shifts up modestly in stress, but BTC/USDC is shifted higher and far more dispersed in the regimes where it exists. On an aggregate basis, this figure supported the conclusion that the USDC-quoted book is materially more price-impacting than the USDT-quoted book.

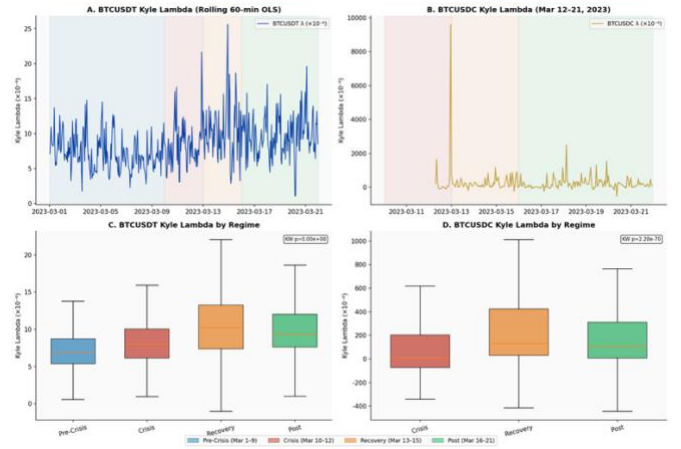


Figure 7: Kyle’s Lambda (price impact) by pair and regime.

Having established a clear quote-currency gap in price impact and illiquidity, we next examine whether the same pattern appears in more direct trading-cost and capacity proxies, namely spread and depth, and how these conditions vary across regimes and venues.

C. Spreads, depth and volatility around the stress window

To connect the microstructure results above to more direct trading-cost and capacity proxies, we summarize a one-minute range-based spread proxy, a volume-based depth proxy, and

realized volatility for BTC/USD and BTC/USDT on Binance.US, segmented by regime. Table X reports levels by regime, while Table XI reports crisis-to-pre-crisis ratios as a simple stress benchmark.

The spread proxy widens sharply during stress. For BTC/USDT, the median spread proxy increases from 3.67 bps pre-crisis to 7.51 bps in crisis (2.05 $\times$), and the 95th percentile increases from 14.80 bps to 26.15 bps (1.77 $\times$). BTC/USD exhibits a nearly identical deterioration, with the median widening from 4.62 bps to 9.31 bps (2.01 $\times$) and the 95th percentile widening from 16.12 bps to 28.30 bps (1.76 $\times$) (Tables X–XI). Interpreted operationally, this means that short-horizon price dispersion and effective trading frictions rise materially in the stress window, which is consistent with the “worse execution when dislocations matter most” narrative developed in the prior sections.

TABLE X

Spread, Depth-Proxy, and Volume by Regime and Quote Currency (Binance.US)

Pair	Regime	N	Spread P50 (bps)	Spread P95 (bps)	Vol P50 (BTC)
BTC/USDT					
	Pre-Crisis	12,960	3.67	14.80	0.6
	Crisis	4,320	7.51	26.15	1.3
	Recovery	4,320	12.41	38.26	3.2
	Post	8,640	9.23	26.31	0.7
BTC/USDC					
	Pre-Crisis	0	—	—	—
	Crisis	1,440	0.00	22.75	0.0
	Recovery	4,320	0.00	29.77	0.0
	Post	8,640	0.00	15.74	0.0
BTC/USD					
	Pre-Crisis	12,960	4.62	16.12	1.8
	Crisis	4,320	9.31	28.30	3.9
	Recovery	4,320	13.72	40.43	6.5
	Post	8,640	11.31	27.78	5.4

Spread = Parkinson high-low estimator $\times 10,000$ (bps). Vol P50 = median BTC volume per 1-min bar. BTC/USDC listed on Binance.US Mar 12; pre-listing rows excluded.

TABLE XI

Stress Ratios (Crisis / Pre-Crisis) by Pair and Metric

Pair	Metric	Pre-Crisis	Crisis	Ratio
BTC/USDT				
	Spread P50 (bps)	3.67	7.51	2.05 $\times$
	Spread P95 (bps)	14.80	26.15	1.77 $\times$
	Volume P50 (BTC/min)	0.56	1.33	2.39 $\times$
	Volume P5 (BTC/min)	0.00	0.02	18.83 $\times$
BTC/USD				
	Spread P50 (bps)	4.62	9.31	2.01 $\times$
	Spread P95 (bps)	16.12	28.30	1.76 $\times$
	Volume P50 (BTC/min)	1.77	3.88	2.19 $\times$
	Volume P5 (BTC/min)	0.07	0.30	4.52 $\times$

Ratio = Crisis value / Pre-Crisis value. BTC/USDC excluded (no pre-listing baseline). Volume P5 = thin-book indicator.

Depth indicators require a slightly more nuanced reading. Median one-minute BTC volume rises in the crisis regime for both BTC/USDT and BTC/USD (Table XI), which is consistent with elevated trading activity under stress. However, the lower tail of volume remains thin in absolute terms and is informative as a “capacity-in-the-worst-minutes” indicator. In other words, stress can bring more flow while still leaving many minutes with limited capacity, which is precisely when price impact and spread widening become binding constraints for execution.

TABLE XII
BTC/USDT Microstructure by Regime (Binance.US)

Regime	N	Spread Med. (bps)	Spread P95 (bps)	Depth Med.	Kyle λ Med.	OBI Mean	Vol. Med.
Pre-Crisis	12,960	2.43	13.68	4,836	6,757,069	-0.055	0.75
Crisis	4,320	6.58	25.99	5,331	6,577,997	+0.004	3.40
Recovery	4,320	11.57	38.25	5,150	9,262,064	+0.016	5.77
Post	8,640	9.33	26.95	4,677	9,605,859	-0.013	4.23

Spread = (ask-bid)/mid $\times 10,000$. Depth proxy = sum of top-5 bid+ask quantity. Kyle λ = price impact per unit volume. OBI = (bid vol. - ask vol.)/(bid vol. + ask vol.). Vol. = trade volume (BTC).

On an aggregate basis, liquidity evidence aligns with the pricing results from the earlier sections. When the unit of account is stable, wedges are small and market quality is relatively benign. When stablecoin credibility is stressed, cross-quote dislocations open at the same time that execution becomes more expensive and capacity becomes more fragile, with systematic differences across quote currencies and confirmation on an offshore venue. The natural next step is persistence: given wider spreads, higher impact, and segmented arbitrage channels, how quickly do dislocations normalize back toward parity, with upcoming legislative endeavours? We address this using the regime-based OU half-life estimates in the next sections, with a series of hypotheticals, and discuss potential implications for trading, risk management, and policy.

VI. REGULATORY OVERLAY

We take this one step further in the next section. Rather than treating stablecoin discounts as a side observation, we interpret them as a unit-of-account shock and ask what a credible U.S. regulatory perimeter would change. We use our regime-based OU half-life estimates, together with simple hypotheticals, to translate confidence shocks into normalization speed and then into implications for trading, risk management, and policy

A. Alterations of Cross-Currency Trading Patterns

The empirical evidence from the March 2023 USDC de-peg demonstrates that when a stablecoin’s peg comes under stress, the law of one price degrades and severe cross-quote dislocations emerge. In periods of unit-of-account uncertainty, the stablecoin FX leg develops a pronounced left-tail discount and BTC/USD and BTC/stablecoin quotes can materially diverge. In our sample, USDC traded materially below par at its trough and cross-quote deviations widened by orders of magnitude, reaching 1,324 bps for BTC/USDC relative to BTC/USD at peak. Regulated stablecoins have the potential to change these trading patterns by reducing the credibility shock that initiates fragmentation, rather than by “forcing” secondary-market prices to print at par.

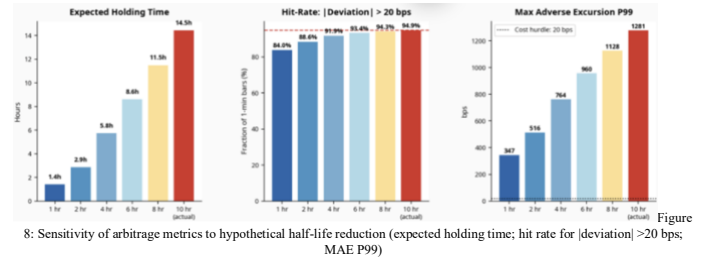
A useful way to frame the link is through the decomposition used throughout our analysis. The total log LOP deviation can be viewed as the sum of a stablecoin FX component, capturing deviations of the stablecoin from par, and a residual component, capturing microstructure and segmentation across venues and quote currencies. In crisis regimes, the FX component reflects a direct unit-of-account shock in which stablecoin credit and liquidity risk are priced into the quote currency. The residual component reflects the market’s

impaired ability to arbitrage and re-anchor prices across fragmented liquidity pools. In our data, these legs do not behave independently in stress: the interaction term dominates variance across regimes and becomes most pronounced during the crisis window, with the venue leg and FX leg moving in a tightly coupled manner. When stablecoins operate without strict regulatory oversight, the risk of impaired redemption and reserve uncertainty is priced into spot markets during stress events and acts as a friction that binds limits to arbitrage. This is consistent with the market structure channel highlighted in our results: cross-venue synchronization in stablecoin discounts indicates a market-wide confidence shock, yet cross-quote and cross-venue dislocations still persist because arbitrage is constrained by capital limits, transfer risks, and venue frictions. In the crisis regime, mean reversion slows sharply and the wedge behaves closer to a near-random-walk, consistent with impaired normalization exactly when dislocations are the largest.

A comprehensive regulatory framework should alter this dynamic by reducing stablecoin credit and liquidity risk and making the instrument closer to a cash-like settlement asset in the states of the world where confidence matters most. By mitigating the unit-of-account distortion at the source, a regulated stablecoin should mechanically compress the FX component of the wedge and indirectly dampen the residual by reducing the liquidity-withdrawal spiral that amplifies segmentation. The appropriate expectation is not “no dislocations,” but smaller tails and faster normalization.

B. *OU counterfactual: how much does faster normalization matter?*

To make the policy link operational, we use the OU framework as a reduced-form yardstick for normalization speed. We start from the regime-specific AR(1) estimates for the residual wedge and map the fitted coefficient into an OU speed κ and half-life. We then run a controlled sensitivity where we hold the crisis distribution of shocks fixed and vary only HL (equivalently κ), recomputing three practical objects: the expected time for a deviation to decay back inside a fixed cost band, the fraction of minutes in which deviation exceeds that band (hit-rate), and a tail risk statistic (P99 max adverse excursion) that summarises how large the position can move against you before it mean-reverts (Fig. 8–9). The reason this matters is that persistence is what turns a “spread trade” into an inventory and funding problem: if dislocations decay in minutes, capital is recycled quickly; if they decay over hours, the same trade carries materially more margin, liquidity, and drawdown risk. This approach mirrors the standard “limits-to-arbitrage” reading in parity settings, where long half-lives are treated as evidence that arbitrage capacity is constrained and deviations can persist even when they look large on paper (e.g., the CIP persistence logic in Du, Tepper and Verdelhan).



8: Sensitivity of arbitrage metrics to hypothetical half-life reduction (expected holding time; hit rate for |deviation| > 20 bps; MAE P99)

Scenario	Half-Life (hrs)	κ (month ⁻¹)	κ (hr ⁻¹)	E(hold) (hrs)	Hit-Rate >20 bps (%)	MAE P99 (bps)
1 hr (GENIUS rule)	1.0	0.01155	99.3	1.44	84.0%	347
2 hr	2.0	0.00578	140.0	2.88	88.6%	516
4 hr	4.0	0.00289	197.6	5.76	91.9%	764
6 hr	6.0	0.00193	242.0	8.64	93.4%	960
8 hr	8.0	0.00144	279.3	11.52	94.3%	1128
10 hr (actual crisis)	10.0	0.00115	312.9	14.46	94.9%	1281

9: Sensitivity Table of Assumptions into OU speed and other Arbitrage Metrics

C. *Alterations of Cross-Currency Trading Patterns*

At the same time, there is a structural tradeoff that matters for interpretation. Regulation can introduce compliance gates that reshape where stablecoins circulate. If regulated stablecoins concentrate on compliant venues while offshore liquidity remains in other instruments, cross-quote arbitrage may still be impeded by jurisdictional segmentation and transfer frictions even if peg risk is substantially reduced. In decomposition terms, regulation compresses the wedge’s FX leg, while a non-trivial residual can remain as a reflection of market structure rather than stablecoin credibility.

D. *Implications of the GENIUS Act and payment-system adoption*

The legislative mechanisms proposed in frameworks like the GENIUS Act, including one-to-one reserve backing in high-quality liquid assets and recurring independent public reserve reporting, matter for market structure because they target the unit-of-account shock directly. By codifying redemption assurance and transparency, such regulation mitigates the basis risk faced by institutional traders who rely on stablecoins for collateral, funding, and settlement. More broadly, a GENIUS-style perimeter pushes stablecoins toward a more numeraire-like role, where stablecoin-quoted markets behave more like conventional currency markets in stress.

The immediate structural implication is better resilience in adverse regimes: smaller tail discounts and faster normalization of cross-quote wedges. This aligns with our findings that the most disruptive episodes combine (i) large parity deviations, (ii) deterioration in liquidity metrics and execution conditions, and (iii) much slower convergence. If transparency and redemption credibility reduce panic-driven runs, the market should see less amplification of price impact and shorter OU half-lives during stress, not merely tighter average parity.

Payment-rail adoption strengthens the importance of this channel. As stablecoins are integrated into settlement workflows, the cost of a unit-of-account shock extends beyond speculative mispricing: it becomes execution risk and operational risk for participants who treat stablecoin quotes as

dollar-equivalent. This is precisely why persistence is policy-relevant: the difference between a 3-minute and a 10-hour half-life is the difference between a transient microstructure disturbance and a prolonged disruption that affects liquidity provision, hedging reliability, and the feasibility of cross-venue convergence.

E. Conclusion and Implications

Overall, our results suggest that stablecoin stress is not just a local “FX” event. It is a unit-of-account shock that propagates into cross-quote pricing and interacts with market structure. During the USDC de-peg, the stablecoin leg moved sharply away from par, but the residual wedge also widened and persistence increased materially. This combination is what makes the episode economically meaningful: dislocations appear exactly when execution conditions deteriorate and normalization slows, consistent with constrained arbitrage and fragmented liquidity.

The trading implication is that profitability is not determined by the size of the wedge alone. Persistence and tail risk matter at least as much. When half-lives are measured in minutes, convergence trades behave like high-frequency spreads. When half-lives extend into hours, the same trade becomes an inventory, funding, and drawdown problem. The OU sensitivity exercise makes this point transparent: faster normalization would compress holding times and reduce adverse excursion risk, even if deviations still open under stress.

For risk management, the main takeaway is that treating stablecoin quotes as dollar-equivalent embeds basis risk and liquidity risk that become state-dependent. In normal regimes, these risks are largely invisible. In stress, they surface through wider spreads, higher price impact, and prolonged deviations. A practical consequence is that exposure limits and hedges should be conditioned on stablecoin stress indicators and on measured persistence, rather than on unconditional volatility alone.

For policy, the evidence supports a targeted interpretation. A GENIUS-style framework is most likely to improve market resilience by shrinking the left tail of stablecoin discounts and speeding up confidence resolution, which should reduce both the magnitude and the duration of cross-quote dislocations. However, our cross-venue results also point to an important limit: even with stronger reserve credibility, fragmentation can persist due to venue segmentation, redemption access, and transfer frictions. In that sense, regulation can compress the unit-of-account channel without fully eliminating residual wedges driven by market structure.

Taken together, the core implication is that stablecoin regulation and settlement adoption should be evaluated less on average parity and more on tail behavior and normalization speed. In our framework, those map directly to the discount distribution and the OU half-life. These are measurable objects that link policy design to execution quality, arbitrage feasibility,

and systemic resilience under stress.

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